



Experimental model identification of open-frame underwater vehicles

Juan P.J. Avila^{a,*}, Décio C. Donha^b, Julio C. Adamowski^b

^a Department of Aerospace Engineering, Federal University of the ABC, Rua Santa Adélia 166, Santo André CEP 09210-170, São Paulo—SP, Brazil

^b Mechanical Engineering Department, University of São Paulo, Av. Prof. Mello Moraes 2231, Butantã CEP 05508-900, São Paulo—SP, Brazil

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ABSTRACT

Most of the works published on hydrodynamic parameter identification of open-frame underwater vehicles focus their attention almost exclusively on good coherence between simulated and measured responses, giving less importance to the determination of “actual values” for hydrodynamic parameters. To gain insight into hydrodynamic parameter experimental identification of open-frame underwater vehicles, an experimental identification procedure is proposed here to determine parameters of uncoupled and coupled models. The identification procedure includes: (i) a prior estimation of actual values of the forces/torques applied to the vehicle, (ii) identification of drag parameters from constant velocity tests and (iii) identification of inertia and coupling parameters from oscillatory tests; at this stage, the estimated values of drag parameter obtained in item (ii) are used. The procedure proposed here was used to identify the hydrodynamic parameters of LAURS—an unmanned underwater vehicle developed at the University of São Paulo. The thruster–thruster and thruster–hull interactions and the advance velocity of the vehicle are shown to have a strong impact on the efficiency of thrusters appended to open-frame underwater vehicles, especially for high advance velocities. Results of tests with excitation in 1-DOF and 3-DOF are reported and discussed, showing the feasibility of the developed procedure.

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1. Introduction

Remotely operated vehicles (ROVs) are being used worldwide for various tasks such as environmental survey, structure inspection, pipeline tracking, mine hunting, heavy work activities, and even for recovering wrecks. The success of ROV complex maneuver, especially near the ocean floor, requires precise motion control. The development of control laws for ROV operations are, as usual, generally based on equations of motion, which, in the case of these vehicles, are strongly dependent on hydrodynamic parameters, showing the importance of an accurate evaluation of these parameters.

A number of methods to identify the hydrodynamic parameters of ROVs have been proposed. Traditionally, hydrodynamic parameters are identified in towing tank tests, using the vehicle itself or its scaled model (Nomoto and Hattori, 1986). Using a Planar Motion Mechanism (PMM), forces and moments are measured in 6-DOF, allowing a complete model identification. However, PMMs are very expensive mechanisms and test procedures are highly time-consuming.

More recently, system identification (SI) approaches have become popular (Caccia et al., 2000; Ridao et al., 2004; Smallwood, 2003; Smallwood and Whitcomb, 2003). This technique is preferable because it makes use of the data of onboard sensors and of the control signals of thrusters in the identification. No other equipment is needed. The technique is cost-effective and the repeatability is high. It is very suitable for the variable configuration of ROVs, in which its payload and shape may change according to the mission.

However, the determination of the hydrodynamic parameters by SI is very difficult and challenging. First, the values of the hydrodynamic parameters are a function of the Keulegan–Carpenter number, which depends on the period and amplitude of the oscillatory motion of the vehicle, as shown by Avila (2008). Second, the determination of the torque/force actually applied to the vehicle by the thrusters is also very difficult, because the effects of thruster–thruster and thruster–hull interactions and of the advance velocity of the vehicle on the thrust produced by thruster must be quantified. Third, the measurements of the sensors are affected by the magnetic field induced by the thruster motors. Moreover, the lack of instrumentation for acceleration measurement, usual in underwater robotics, and the low sampling rate of the instruments frequently used to measure the velocity of the vehicle (for example, the Doppler sonar) impose further heavy restrictions on identification.

* Corresponding author. Tel.: +55 11 49960113; fax: +55 11 49960089.

E-mail addresses: juan.avila@ufabc.edu.br (J.P.J. Avila), decidonha@usp.br (D.C. Donha), jcadamow@usp.br (J.C. Adamowski).

Few hydrodynamic parameter identification methodologies for underwater vehicles have been proposed in the literature. Caccia et al. (2000) proposed an off-line identification based on the least-squares method (LS) and used the data of position provided by both digital altimeter and compass to identify the hydrodynamic parameters of a ROV. Smallwood and Whitcomb (2003) presented an on-line adaptive identification method and used a Sonic High Accuracy Ranging and Positioning System (SHARPS) to measure the position of a ROV. However, the measurement of the position of a vehicle by SHARPS is still expensive. Recently, vision-based localization has become very popular, since it can provide precise location data and, perhaps more importantly, is low-cost. Parameter identification methods including a vision-based navigation system to estimate the position of the vehicle was proposed by Ridao et al. (2004) and Chen et al. (2007). Yet, the navigation of ROVs based on cameras is not standard and navigation algorithms are still a little complex for application.

Most of the published works about hydrodynamic parameter identification of underwater vehicles, including the works mentioned above, estimate the values of the parameters by means of free run trials in a single DOF. During those tests, thrusters apply torque/force at a single DOF, producing decoupled motions. Moreover, no works were found considering the data acquired by DVLs (Doppler Velocity Logs) in the identification process, although DVL sensors are usually available in underwater vehicle navigation.

Also, it is well known that the thrusters working in open-frame underwater vehicles lose efficiency due to the interaction with other thrusters or with the vehicle structure (Caccia et al., 2000; Goheen and Jeffereys, 1990). Thruster–hull interactions are a well-known and studied phenomenon in surface vessels (Newman, 1977); however, for underwater vehicles, they have seldom been taken into account. A thruster is said to have unity efficiency when the thrust it produces is equal in value to the thrust produced in open-water tests and at bollard-pull conditions. A single work published on hydrodynamic parameter identification of ROVs includes the effects of thruster–thruster and thruster–hull interactions on the dynamic model of the vehicle (Caccia et al., 2000). Depending on the structural configuration of the ROV, the thrust reduction due to thruster–thruster and thruster–hull interactions is considerable and will significantly affect the precision of the estimated values of the hydrodynamic parameters. Caccia et al. (2000) determined a thrust reduction equivalent to 40% of the value of the thrust produced by the thrusters at bollard-pull tests, in which case the ROV pulls up in heave. A simple mathematical model was developed herein for the actual thrust force applied to the vehicle, and the parameters of the model were identified through tests with the vehicle in bollard-pull, differently from Caccia et al. (2000), who simultaneously identified the parameters of the thruster model and the parameters of the vehicle from free run tests. The developed thrust model also includes the efficiency reduction of the thrusters due to the advance velocity of the vehicle.

This work presents the final results of a research focusing on the development of a procedure for the modeling and hydrodynamic parameter identification of an open-frame ROV using on-board sensor data. Differently from most published works, it considers: (i) the velocity data of the vehicle provided by a DVL, (ii) the parameter identification of a coupled dynamic model, and (iii) the identification of drag parameters from constant velocity tests and the utilization of these values to determine the inertia and coupling parameters. The proposed identification procedure is explained in Section 5 and basically consists of the following. First, the efficiency coefficients of the system of thrusters are experimentally identified from bollard-pull tests. The efficiency coefficients are used to compute the torque/force actually applied upon the vehicle during its motion, taking into account the loss of

thrust due to the thruster–thruster and thruster–hull interactions and to the velocity of advance of the vehicle. Subsequently, the drag parameters of the vehicle are determined from constant velocity tests in 1-DOF and, finally, using the information of the drag parameters, the inertia and coupling parameters are identified in tests of variable velocity. The idea of first identifying the drag parameters and later using them to identify the inertia parameters was adopted, first by Hwang (1980) and Abkowitz (1980) in the case of a ship, and by Caccia et al. (2000) in the case of open-frame underwater vehicles.

It is verified here that this identification scheme allows obtaining values for the hydrodynamic parameters consistently with the physical conception of the vehicle. Moreover, the uncertainty in the hydrodynamic parameter estimated values decreases, once the length of the vector of unknown parameter is reduced.

The parameter identification method used in this work is the classical least-squares algorithm applied to the integral form of the system dynamic equations (Ridao et al., 2004). This method is convenient to identify parameters of coupled models of ROVs since the majority of the state variables of these vehicles are usually known (measured by the sensors) and, due to the relatively slow dynamics of these vehicles (increasing the signal–noise ratio).

The proposed methodology was applied to model and to identify LAURS, a ROV prototype developed in the Laboratory of Sensors Actuators at the University of São Paulo for robotics research and deep water technological applications. Some of the results obtained using the proposed procedure are compared with the results obtained by tests of LAURS prototype in a towing tank using a PMM, see Avila (2008) for comparison and discussion of results.

The paper is organized as follows. Section 2 describes the mechanical design, the thrusters and sensor systems of the LAURS vehicle. Modeling of the ROV dynamics and the parameter identification method are discussed in Section 3. In Section 4, the model developed to estimate the actual thrust applied by the thrusters is presented. Experimental results obtained by adapting this procedure to the identification of the LAURS vehicle are outlined in Section 5. Finally, Section 6 presents some concluding remarks.

2. The LAURS design

2.1. Mechanical design and sensor system

The LAURS is defined as an open-frame ROV and was initially conceived to provide inspection and intervention capabilities in

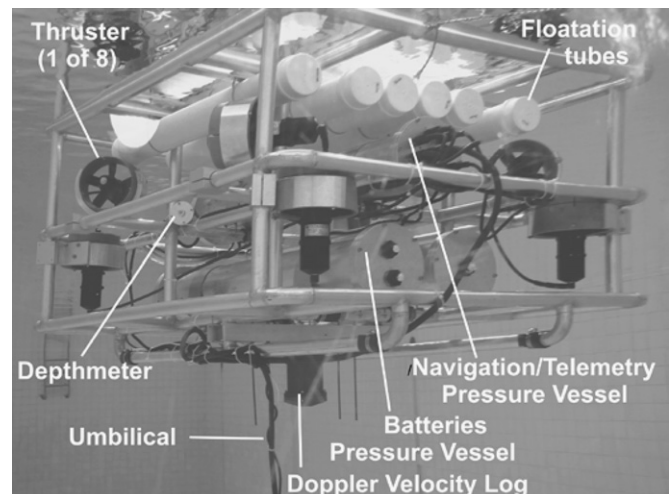


Fig. 1. Physical layout of the LAURS.

specific missions in deep-water oil fields. The main feature of this underwater vehicle is an interchangeable tool-sled design. Despite the use of the remotely operated term, the LAURS was conceived as having semi-autonomous behavior, i.e., the vehicle can be remotely operated but has an autonomous mode that can be used to approach targets.

The vehicle configuration is composed of an aluminum tubular structure, 1.4 m long $\times$ 1.2 m wide $\times$ 0.9 m high, equipped with three pressure vessels of the same dimensions, 1.0 m long $\times$ 0.17 m diameter, see Fig. 1. Its mass is 189 kg and the weight–buoyancy force is positive 35 N. Actuation is provided by eight DC brushless electric thrusters. The LAURS is intrinsically stable in pitch and roll and divided for convenience into two parts: upper and bottom. The upper part of the vehicle contains a layer of PVC tubes for buoyancy properties, a pressure vessel for the electronics and sensors, and the four horizontal thrusters. The bottom part of the vehicle consists of two pressure vessels that contain batteries and four vertical thrusters. Modular structural components allow LAURS to be easily reconfigured in agreement with specific tasks. The overall structure of the vehicle is symmetric with respect to both the XZ and YZ planes, being the X and Y axis parallel and perpendicular to the axis of the cylinders, respectively.

The LAURS is powered by an isolated 4.5 kW DC power supply (through an umbilical) and its control architecture is based on an on-board computer system running a VxWorks real time operating system. The computer system is devoted to sensor data logging, guidance and motion control. The XYZ position and heading of the vehicle are actively controlled by a closed-loop control system.

The sensor system of LAURS is composed of a set of sensors which permits the full 6-DOF position measurement. The XYZ velocity is measured using a DVL; see Fig. 2. This velocity data is expressed in an inertial reference frame (Fossen, 1994) by using the attitude and heading data also provides by DVL. Subsequently, the inertial coordinate velocity is integrated numerically by using the fourth order Runge–Kutta integration scheme to obtain the linear position of the vehicle. Depth measurement is achieved via an analog pressure transducer. Heading and yaw velocity are measured using a digital compass and a fiber optic gyro, respectively. Two electrolytic tilt sensors measure the roll and



Fig. 2. LAURS DVL sensor.

pitch of the vehicle. The altitude of the vehicle is measured using a digital altimeter. The entire sensor suite of the vehicle is listed in Table 1.

2.2. Propulsion system

The LAURS propulsion system is composed of eight Tecnydyne Model 1020 DC brushless thrusters with velocity feedback loop. Avila (2008) verified that the settling time of thruster step response as well as the settling time of LAURS response to surge step input have a time constant equal to 0.1 s and 3.0 s, respectively. Based on this information, it was assumed that the vehicle parameter identification is not affected by the dynamics of the thruster time delays, which were therefore omitted. Considering this, the following thruster model was used to calculate the axial force, F , produced by the propeller:

$$F = aV^2 + bV, \quad (1)$$

where V is the control voltage, which is applied to the thruster servo-amplifier, and a and b are force coefficients of a typical thruster. Coefficients a and b of Eq. (1) were identified experimentally by putting the whole actuator in a water tank at bollard-pull conditions and measuring the force as a function of a set of input voltages. The least-squares method was used to identify a and b . This procedure was repeated with each of the eight thrusters of the vehicle. For example, the estimated values of the force coefficients for the thruster identified as Thruster One are $a = 11.53 \pm 0.1$ [N/V²] and $b = 4.21 \pm 0.36$ [N/V] for positive thrust, and $a = 7.87 \pm 0.15$ [N/V²] and $b = 4.68 \pm 0.55$ [N/V] for negative thrust.

3. Parameter identification method

3.1. Dynamic modeling

Let us define the vector $\boldsymbol{\eta} = [\boldsymbol{\eta}_1^T \ \boldsymbol{\eta}_2^T]^T$, where $\boldsymbol{\eta}_1 = [x \ y \ z]^T$ is the vector of vehicle position coordinates in an earth-fixed reference frame and $\boldsymbol{\eta}_2 = [\phi \ \theta \ \psi]^T$ is the vector of vehicle Euler-angle coordinates in the earth-fixed reference frame, and the vector $\mathbf{v} = [\mathbf{v}_1^T \ \mathbf{v}_2^T]^T$, where $\mathbf{v}_1 = [u \ v \ w]^T$ is the vector of vehicle linear velocity expressed in the vehicle-fixed reference frame and $\mathbf{v}_2 = [p \ q \ r]^T$ is the vector of vehicle angular velocity expressed in the vehicle-fixed reference frame. In Fig. 3, the defined frames and the elementary vehicle motions are illustrated.

The equations of motion of an ROV can be written in the vehicle-fixed reference frame in the form (Abkowitz, 1969; Fossen, 1994):

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{g}(\boldsymbol{\eta}_2) + \mathbf{b} = \boldsymbol{\tau}, \quad (2)$$

where

$\mathbf{M}$ mass matrix that includes both rigid body mass and added mass,

Table 1
LAURS experimental instrumentation.

Variable	Sensor (manufacturer)	Precision, update rate	Output
Heading	Compass TCM2-50 (PNI)	$\pm 1^\circ$, 13 Hz	Digital
Roll and pitch	Tilt Series 757 (Applied Geomechanics)	$\pm 1^\circ$, 20 Hz	Analog
Depth	Pressure sensor MPX5100DP (Motorola)	50 mm, 20 Hz	Analog
Yaw rate	Fiber optic gyro, E-Core2000 (KVH)	Bias $< 2^\circ \text{ h}^{-1}$, 9 Hz	Digital
Height	Altimeter, PA200 (Tritech)	1 mm, 10 Hz	Digital
XYZ velocity, roll, pitch, yaw, height	Doppler velocity log (NavQuest 600 Micro)	$0.2\% \pm 1 \text{ mm/s}$, 3 Hz	Digital

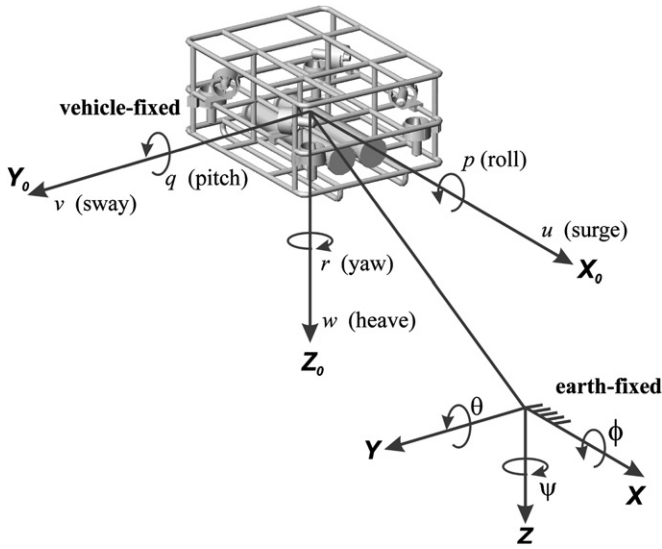


Fig. 3. Reference frames and elementary motions of the vehicle.

$C(v)$ coriolis and centripetal matrix,
 $D(v)$ hydrodynamic damping matrix,
 $g(\eta_2)$ vector of gravitational and buoyant forces,
 b vector of forces/moments due to umbilical cable,
 τ vector of control forces and moments generated by thrusters.

The hydrodynamic coefficients of Eq. (1) can be identified through towing tank tests of the vehicle itself or a reduced scale model, such as was mentioned in the introduction of this work. Special equipment called planar motion mechanism is built above the towing tank to move the vehicle in a planar motion. But, although towing tank tests allow identifying all the hydrodynamic coefficients of the vehicle, these tests are complex, expensive and highly time-consuming.

In the last decade, the scientific community has preferred to carry out tests with the propelled prototype as well as to utilize both data of onboard sensors and control signals of thrusters in the identification. This technique is cost-effective and repeatable. Yet, due to the difficulty of controlling the vehicle at 6-DOF, as required by this technique, 1-DOF experimental runs have been preferred and, consequently, 1-DOF models have been frequently used for identification. On the other hand, experimental runs with excitation in many DOFs increase the uncertainty of the estimated parameter values because of (1) the effects of the thruster–thruster and thruster–hull interactions on measured forces and (2) the difficulty in determining the force/torque really applied to the vehicle under control. Favorably, 1-DOF ROV models are preferred for the purposes of controller design specially for ROVs that change in configuration according to their mission, since it is simpler and cheaper to design a controller based on 1-DOF model.

Usually ROVs have two symmetry planes in its geometrical configuration as well as in its mass distribution. Other characteristics of the ROVs are its slow velocity and modest attitude changes typical of this class of underwater vehicles. Therefore, taking into account the characteristics above given the motion of a ROV can be uncoupled and a 1-DOF dynamic model can be used to model the principal motions of this type of vehicles. This approximation has been commonly adopted by the scientific community (Caccia et al., 2000; Smallwood and Whitcomb, 2003; Ridao et al., 2004). This 1-DOF model is obtained from Eq. (2) by neglecting off-diagonal entries and coupling terms, tether dynamics, as well as assuming a constant added mass. This approximation relies on the fact that: (1) the off-diagonal

elements of the added mass matrix of a rigid body having three symmetry planes are identically null (Newman, 1977); (2) the off-diagonal elements of the positive definite matrix are much smaller than their diagonal counterparts (Fossen, 1994); and (3) the hydrodynamic damping coupling is negligible at low speeds. The resulting model structure is

$$m_i \dot{x}_i + d_{L,i} x + d_{Q,i} |x| + b_i = \tau_i \quad (3)$$

where, for each DOF i , x_i is the velocity at the direction of the DOF i ; m_i is the effective mass; $d_{L,i}$ and $d_{Q,i}$ are the linear and quadratic drag coefficients, respectively; b_i is the modeling of the cable effects; τ_i is the net control force. We also use the 1-DOF model given by Eq. (3) here and go beyond, identified a 3-DOF model as will be discussed in Section 5.4.

3.2. Hydrodynamic parameter identification

The parameter identification method used in this work is the classic least-squares algorithm (LS) applied to the integral form of the system dynamic equations. In order to explain the identification numerical procedure, the identification of the parameters of the following non-linear and coupled differential equation will be carried out:

$$\dot{x}(t) = \alpha x(t)y(t) + \beta x(t) + \gamma x(t)|x(t)| + \delta \tau(t) + \zeta, \quad (4)$$

where $x(t)$ and $y(t)$ are the experimentally measured velocities of the vehicle in two different DOFs, which are considered to be known. The acceleration of the vehicle, $\dot{x}(t)$, is unknown while the torque/force applied, $\tau(t)$, is known. The coefficients α , β , γ , δ and ζ are the parameters to be identified. It is worth stressing that Eq. (4) has not been derived from Eq. (2). Eq. (4) is merely a mathematical expression introduced in this work to show the feasibility of the identification numerical method. Actually, Eq. (4) has been obtained by adding a non-linear term, $\alpha x(t)y(t)$, to the 1-DOF model given by Eq. (3).

Instrumentation to measure accelerations is not standard in most underwater vehicles because accelerometers are highly sensitive to mechanic vibrations. The usual creeping motion of ROVs, as well as the mechanic vibrations induced by thrusters, hinder the use of accelerometers. On the other hand, calculation of acceleration via direct numerical differentiation, an acasual operation, is easily corrupted by sensor noise. The need of a parameter identification algorithm that does not consider the acceleration as input data is then clear.

The integration of Eq. (4) allows avoiding the use of acceleration input data in the parameter identification algorithm. Integrating Eq. (4) in the time interval t_{k-1} and t_k leads to the following equation:

$$x(t_k) - x(t_{k-1}) = H_k \theta \quad (5)$$

where

$$H_k = \begin{bmatrix} a_k & b_k & c_k & d_k & e_k \end{bmatrix} \quad \text{with}$$

$$a_k = \int_{t_{k-1}}^{t_k} x(s)y(s)ds = \frac{1}{2}[x(t_{k-1})y(t_{k-1}) + x(t_k)y(t_k)] \Delta t_k,$$

$$b_k = \int_{t_{k-1}}^{t_k} x(s)ds = \frac{1}{2}[x(t_{k-1}) + x(t_k)] \Delta t_k,$$

$$c_k = \int_{t_{k-1}}^{t_k} x(s)|x(s)|ds = \frac{1}{2}[x(t_{k-1})|x(t_{k-1})| + x(t_k)|x(t_k)|] \Delta t_k,$$

$$d_k = \int_{t_{k-1}}^{t_k} \tau(s)ds = \frac{1}{2}[\tau(t_{k-1}) + \tau(t_k)] \Delta t_k,$$

$$e_k = \int_{t_{k-1}}^{t_k} 1ds = \Delta t_k$$

and

$$\Delta t_k = t_k - t_{k-1},$$

and θ is the parameter vector given by:

$$\theta = [\alpha \quad \beta \quad \gamma \quad \delta \quad \zeta]^T.$$

Eq. (5) can be re-written in the usual predictor form as follows:

$$\hat{x}(t_k) = \tilde{x}(t_{k-1}) + H_k \theta \quad (6)$$

where $\hat{x}(t_k)$ denotes the one-step ahead prediction of the output and $\tilde{x}(t_{k-1})$ is the state of the system known from experimental measurements.

Prediction error $\varepsilon(t_k)$ is defined as follows:

$$\varepsilon(t_k) = \tilde{x}(t_k) - \hat{x}(t_k) \quad (7)$$

By substituting Eq. (6) into Eq. (7) and reordering the equation results in:

$$\tilde{x}(t_k) - \tilde{x}(t_{k-1}) = H_k \theta + \varepsilon(t_k) \quad (8)$$

Applying Eq. (8) to the sampled data in N -instants, the following system of algebraic equations is yielded:

$$\underbrace{\begin{bmatrix} \tilde{x}(t_2) - \tilde{x}(t_1) \\ \tilde{x}(t_3) - \tilde{x}(t_2) \\ \vdots \\ \tilde{x}(t_k) - \tilde{x}(t_{k-1}) \\ \vdots \\ \tilde{x}(t_N) - \tilde{x}(t_{N-1}) \end{bmatrix}}_Y = \underbrace{\begin{bmatrix} H_1 \\ H_2 \\ \vdots \\ H_k \\ \vdots \\ H_N \end{bmatrix}}_H \underbrace{\begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \\ \zeta \end{bmatrix}}_{\theta} + \underbrace{\begin{bmatrix} \varepsilon(t_1) \\ \varepsilon(t_2) \\ \vdots \\ \varepsilon(t_k) \\ \vdots \\ \varepsilon(t_N) \end{bmatrix}}_{\varepsilon} \quad (9)$$

Eq. (9) can be re-written as

$$Y = H\theta + \varepsilon, \quad (10)$$

where Y is the measurement vector, in which each element is given by $y(t_k) = \tilde{x}(t_k) - \tilde{x}(t_{k-1})$ and H is the regression matrix. Notice that Eq. (10) is linear in relation to θ and thus θ can be conveniently obtained by using the LS algorithm.

The identification of θ is achieved through the minimization of the following quadratic scalar cost function:

$$J = (Y - H\theta)^T (Y - H\theta), \quad (11)$$

or equivalently

$$J = \sum_{k=1}^N [y(t_k) - H_k \theta]^2 \quad (12)$$

where J is the quadratic sum of errors defined in Eq. (9). The LS estimator, $\hat{\theta}_{LS}$, that minimizes J is given by Goodwin and Payne (1977):

$$\hat{\theta}_{LS} = (H^T H)^{-1} H^T Y \quad (13)$$

Assuming that the components $\varepsilon(t_k)$ of the noise vector are zero mean, constant, non-correlated and additives, the estimator defined in Eq. (13) is an efficient, unbiased and an estimator of minimum variance (Beck, 1977) and therefore $\hat{\theta}_{LS}$ is an optimal estimator.

Following the LS theory, the standard deviation of the estimated parameters is computed as:

$$\sigma_{LS} = \sqrt{\text{diag}((H^T H)^{-1} \sigma^2)} \quad (14)$$

where σ is the Gaussian zero mean measurement noise variance. As suggested by Goodwin and Payne (1977), if the variance is unknown, it can be estimated by

$$\hat{\sigma}^2 = \frac{(Y - H\hat{\theta}_{LS})^T (Y - H\hat{\theta}_{LS})}{n-p}, \quad (15)$$

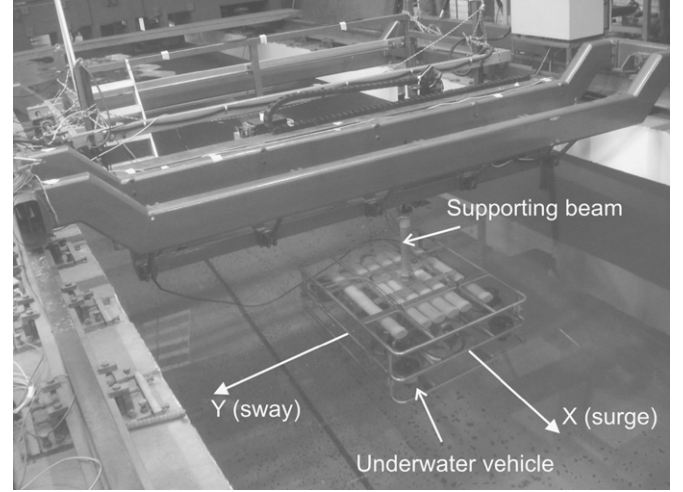


Fig. 4. The LAURS during the bollard-pull tests in water tank.

where $\hat{\sigma}$ is an estimated value of the variance σ and n and p are the length of vectors Y and θ , respectively.

In the computation of H_k , defined in Eq. (5), some numerical integration error is introduced owing to the low sampling rate of the DVL sensor (3 Hz). However, the authors concluded that no significant error is introduced due to the low sampling rate for the DVL, because of the slow dynamics of the vehicle (a system with relatively high time constant). The response of the vehicle to a step input has a time constant equal to 3 s, for the surge direction.

4. Estimation of efficiency of the thruster system

Here, the following model is used to estimate the actual force applied by the thrusters upon the vehicle in operation:

$$\tau = \eta_{inter} \eta_{adv} \tau^{nom}, \quad (16)$$

where τ is the actual force, τ^{nom} is the bollard-pull thrust measured in open-water, here named “nominal” thrust and calculated by Eq. (1); η_{inter} and η_{adv} are efficiency coefficients of the thruster system in which first the losses of thrust due to the thruster–thruster and thruster–hull interactions and second, the loss due to the vehicle advance velocity are taken into account.

The η_{inter} coefficient was experimentally identified by tests of LAURS in the water tank at bollard-pull conditions and, η_{adv} was calculated from the experimental data provided by the thruster manufacturer.

4.1. Identification of η_{inter}

Multiple tests in surge, sway and heave DOFs, one DOF at a time, were conducted with the LAURS in bollard-pull conditions in the towing tank of the Technological Research Institute of São Paulo (IPT). During the tests, different sinusoidal thrust profiles of varying magnitude and frequency were applied, while actual thrust forces acting on the vehicle were measured using load cells. The sinusoidal thrust profile was generated by applying control voltages calculated using Eq. (1) to the thrusters. Fig. 4 shows the LAURS during the bollard-pull tests at surge.

During the bollard-pull tests in surge, the thrust force was applied by horizontal thrusters T5, T6, T2 and T8, when T5 and T6 rotated clockwise while T2 and T8 rotated anticlockwise. Fig. 5 shows the arrangement of the thrusters. Fig. 6 shows a typical surge experiment conducted to measure the actual force acting on the vehicle, by means of load cells, when the following nominal

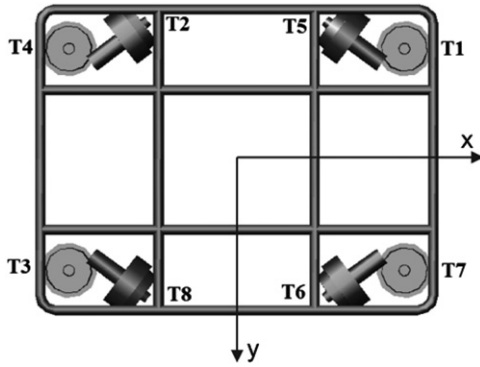


Fig. 5. Configuration of thrusters. Horizontal thrusters: T5, T6, T2 and T8. Vertical thrusters: T1, T7, T4 and T3.

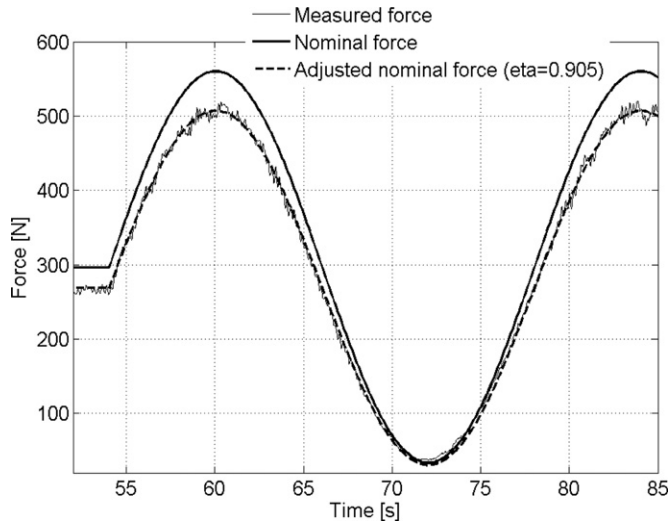


Fig. 6. Efficiency reduction of the T5, T6, T2 and T8 horizontal thrusters due to thruster–thruster and thruster–hull interactions.

force input (dotted line) was applied:

$$\tau_u^{nom} = 263.27 \sin(2\pi t/24) + 296.80 \quad [\text{N}],$$

where t is the time.

Coefficient η_{inter} was obtained by minimizing the difference (in the sense of the least-squares) of the measured force, τ , and the adjusted nominal force, $\eta_{inter} \tau_u^{nom}$. The value calculated for η_{inter} was 0.905.

Table 2 lists the estimated values for η_{inter} in surge, sway and heave directions. It is observed that when only thrusters T5 and T6 were in use in surge, the estimated value of η_{inter} was almost unitary (0.97), indicating that thruster–thruster and thruster–hull interactions can be disregarded.

4.2. Determination of η_{adv}

In principle, coefficient η_{adv} should be obtained by towing the vehicle with a predetermined constant velocity while different step force inputs are applied to the vehicle. The data obtained from this experiment would be the advance velocity (equal to the towing velocity) and values of the constant force applied to the vehicle upon thrusters and measured by load cells. However, since it was not possible to execute these experiments, a model for η_{adv} was deduced based on the experimental data provided by the manufacturer of the thrusters as follows:

$$^*\eta_{adv} = -0.1946v_a + 0.9967, \quad (17)$$

Table 2

Estimated values for the coefficient η_{inter} .

DOF	Thruster applying force	η_{inter}
Surge	T5–T6	0.97
Surge	T5–T6 and T2–T8	0.905
Sway	T6–T8 and T5–T2	0.90
Heave (down)	T1–T3 and T4–T7	0.93

where the symbol * indicates the reference to only a thruster and v_a is the advance velocity of the thruster. v_a is considered to be equal to the velocity of advance of the vehicle. Eq. (17) is considered to be valid only for velocities in the range of 0–1 m/s.

5. Experimental results

In this Section, results of hydrodynamic parameter identification tests of LAURS are presented and analyzed. Tests were conducted in the swimming-pool of the campus of the University of São Paulo in June 2008. During the experimental trials, the vehicle was powered through an umbilical cable by a 4.5 kW DC power source. The buoyancy force was 33.4 N upwards.

The proposed identification procedure to obtain the hydrodynamic parameters of LAURS in one DOF was as follows. First, bollard-pull tests were carried with the vehicle to quantify the loss of thrust due to the thruster–thruster and thruster–hull interactions. Subsequently, constant velocity tests were conducted to determine the drag coefficients. Finally, harmonic oscillatory tests were conducted to determine the added inertia/mass as well as the terms that take into account the hydrodynamic coupling. In the last step, a number of oscillatory tests with different amplitudes and periods of oscillation were conducted to analyze the variation of the values of the added inertia/mass with the Keulegan–Carpenter number. Two characteristics of the identification procedure outlined above stand out. First, the drag parameters are determined from constant velocity tests, and, subsequently, based on the estimated values of the drag parameters, the added inertia/mass parameters are determined from oscillatory tests. Second, the loss of thrust of the thrusters when they operate near the ROV is discounted from the total thrust in open waters, to take into account the actual thrust force applied to the vehicle. The actual thrust force applied to the vehicle is calculated using Eqs. (16) and (17) and the data in Table 2.

Tests conducted in surge, heave and yaw DOFs as well as the results of the maneuvers in horizontal plane are presented and discussed as follows.

5.1. Surge identification

Multiple tests of constant and variable velocity in surge direction were conducted with LAURS. In all the experiments, the force input was applied from thrusters T5 and T6, while the heading was controlled by thrusters T2 and T8; see Fig. 5. Since the thrust force is applied from thrusters T5 and T6 (located at the front of the vehicle), no loss of thrust due to thruster–thruster and thruster–hull interactions (i.e. $\eta_{inter} = 1$, see Table 2) is assumed, but actually it does exist due to the advance velocity. In relation to the control torque applied to the vehicle, the heading control system of LAURS was programmed to apply relatively low torque levels as from thrusters T2 and T8. Low torque levels were necessary so as not to significantly alter the thrust profile required for the identification.

The constant velocity tests were carried out by applying a step force input to the vehicle at 18 s, enough time to guarantee

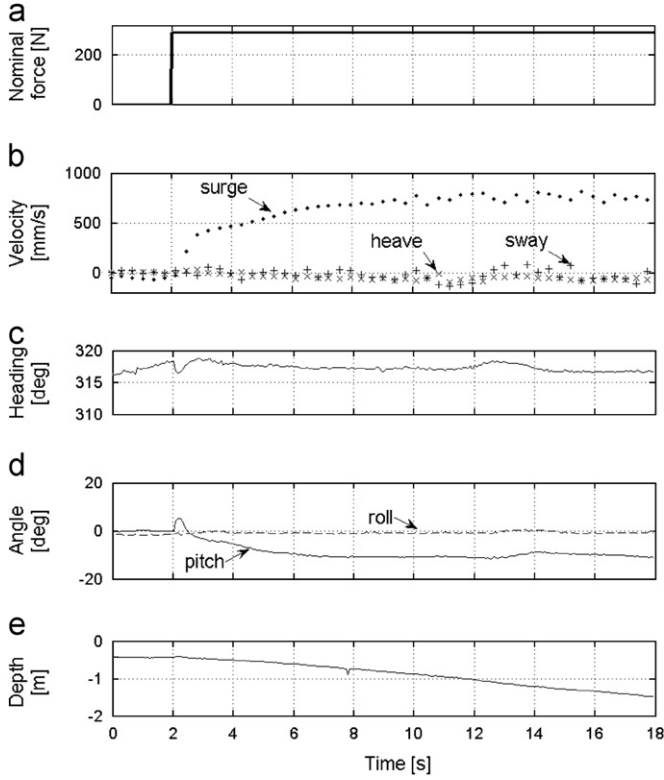


Fig. 7. Surge DOF: experimental data for constant velocity test. (a) Nominal force, (b) surge, sway and heave velocities measured using DVL, (c) heading provided by compass, (d) roll and pitch angles measured by inclinometers and (e) depth measured using pressure sensor.

a stationary condition. Fig. 7 shows the data acquired at one of the constant velocity tests. In Fig. 7b, the surge, sway and heave velocities are shown. The sway and heave velocities are observed to be insignificant as compared to the surge component value, which is equal to 0.748 ± 0.045 m/s (mean value). The requirement of constant heading for the vehicle was satisfied, as shown in Fig. 7c. In this experiment, the motion of the vehicle is observed to be similar to the motion of a body on an inclined plane with slope equal to the pitch angle of the vehicle. This observation is based on the results presented in Fig. 7d and e that respectively show constant pitch angle of mean value equal to -10.25° and a linear variation in depth.

Experimentally, it was verified that the surge motion of the vehicle in the velocity range of 0–0.47 m/s does not present a significantly coupling with the pitch motion. Within this velocity range, the registered maximum pitch angle for the vehicle was of -2.63° , see Fig. 9b. However, for velocities above 0.5 m/s, a relatively high coupling between both surge and pitch axes was observed. The vehicle advances forward with a constant pitch angle. The value of this pitch angle can be determined from the equilibrium of moments about the gravity center of the vehicle. In the equilibrium analysis, the moments produced by thrust, drag and buoyancy forces should be considered. Fig. 9b shows that the pitch angle of the vehicle reaches a value of -12° at a surge velocity of 0.81 m/s. Note that, in most publications on underwater vehicle parameter identification, the tested advance velocities do not exceed the value of 0.4 m/s (Caccia et al., 2000; Ridao et al., 2004; Smallwood and Whitcomb, 2003), and for this reason, they do not report any axis coupling results.

The mathematical model of LAURS surge motion at stationary conditions is given by

$$\tau_u = k_{u|u}|u|u| + k_u u + b, \quad (18)$$

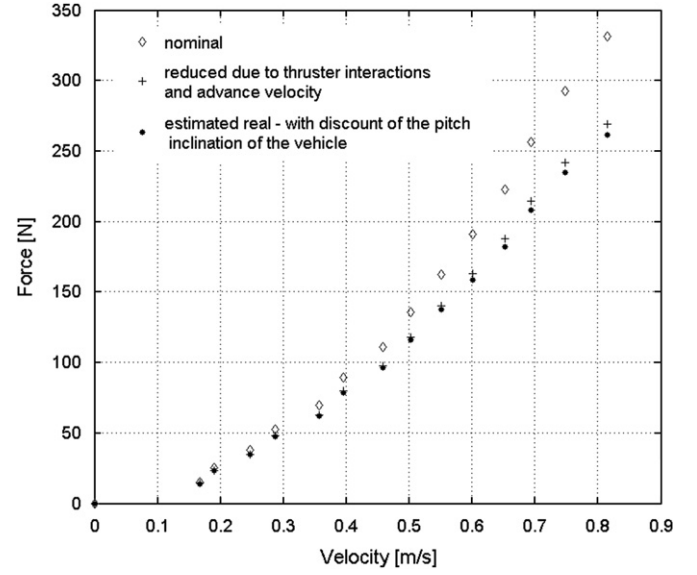


Fig. 8. Surge DOF: nominal thrust force vs. actual thrust force for the experiments of constant velocity.

where u is the surge velocity, τ_u is the actual force applied, $k_{u|u}|$ and k_u are the quadratic and linear drag parameters, respectively, and b a disturbance. An expression for the estimation of τ_u may be as follows:

$$\tau_u = \eta_{adv} \eta_{inter} \tau_u^{nom} + (W - B) \sin(|\theta|), \quad (19)$$

where $\eta_{inter} = 1$ and η_{adv} are efficiency coefficients of thruster defined in Eqs. (16) and (17), respectively, τ_u^{nom} is the applied nominal force calculated by using Eq. (1), W and B are the weight and the buoyancy force upon the vehicle, respectively, and θ is the pitch angle of the vehicle. Since the vehicle at surge motion has a negative pitch angle, we take the absolute value of this angle in order to satisfy the equilibrium of forces given by Eq. (19). The first term on the right-hand side of Eq. (19) is the liquid thrust force obtained by subtracting the effects of the thruster–thruster and thruster–hull interactions and the advance velocity of the vehicle. The second term on the right-hand side of Eq. (19) takes into account the effects of the vehicle pitch. A comparison between the nominal and actual force applied upon the vehicle is shown in Fig. 8.

Fig. 9a shows experimental results, in which drag curves obtained through tests with self-propulsion and by towing the vehicle in the water tank are compared. Results are observed to be relatively well agree, but a small reduction of about 9% (mean value) for the thrust force in relation to the measured force in the towing tank is observed. It is possible to conclude that such thrust reduction is mainly due to the velocity of the water going into the propeller (which coincides with the advance velocity of the vehicle) and neither to the thruster–thruster interaction, since only the front thrusters T5 and T6 (see Fig. 5) push the vehicle, nor to the thruster–hull interaction, once the design of the vehicle prevents it.

The actual thrust force is calculated using Eq. (16) on the base of calculus of the coefficient η_{adv} given by Eq. (17). The latter equation was provided by the manufacturer of the thruster and obtained by towing the thruster with its axis parallel to the towing direction. Error in the estimation of the thrust force using Eq. (17) occurs because Eq. (17) does not consider any angle between the axis of the thruster and the advance direction of the thruster. Actually, that the axis of the front thrusters forms a 36° angle with the longitudinal axis of the vehicle.

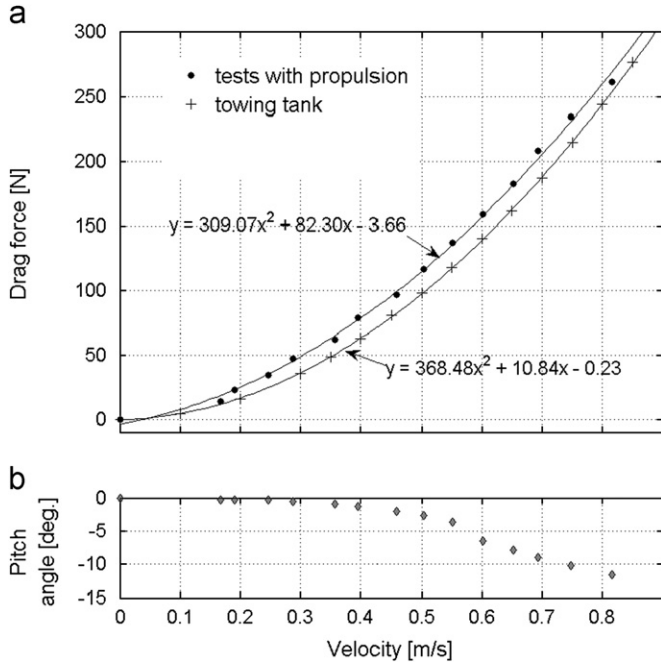


Fig. 9. (a) Comparison of drag curves for surge motion of the LAURS obtained through experiment and (b) pitch angle.

Table 3
Surge DOF: drag parameters for the LAURS.

Experimental approach	$k_{u u }$ (kg/m)	k_u (kg/s)	b (N)	J (N ²)
Towing	368.48 ± 3.81	10.84 ± 3.48	-0.23 ± 0.71	0.66
Propelled	309.07 ± 18.80	82.30 ± 16.61	-3.66 ± 3.26	12.10

The drag parameters of Eq. (18) were identified from the force and velocity data shown in Fig. 9 by using the LS method. Table 3 shows the estimated values of surge drag parameters of LAURS.

The surge model considered in this work is given by

$$m_u \ddot{u} = -X_{\dot{q}} \dot{q} - k_u u - k_{u|u|} |u| |u| + \tau_u - b, \quad (20)$$

where the left-hand side of Eq. (20) is the force induced by the virtual mass m_u and $\ddot{u}$ is the surge acceleration of the vehicle. The first term on the right-hand side of Eq. (20) is the force in surge induced by pitch acceleration $\dot{q}$, where $X_{\dot{q}}$ is the added mass in surge due to $\dot{q}$.

In this work, we propose a ROV hydrodynamic parameter identification procedure. A characteristic of this identification procedure is that this satisfies the requirement of identifiability of model parameters, i.e. “actual values” for the hydrodynamic parameters are obtained. We understand the meaning of the “identifiability” word in accordance with the definition given by Hwang (1980). A parametric model of a system is parameter identifiable if the parameters can be determined uniquely from the input–output relation.

We initially tried to simultaneously identify all the parameters of Eq. (20) using a single experimental datum, but two problems were found: (1) some of the parameters take absurd values which are inconsistent with the conception of the vehicle and (2) the parameters take significantly different values by changing both the length of the data window and degree of the polynomial filter. Paradoxically, despite the existence of the aforementioned problems, the simulated response agreed very well with that the

experimentally measured. Therefore, a clear problem of parameter identifiability arises and the authors verified that this derives from the DVL noise. The DVL noise level caused the ill-conditioning of the matrix $(H^T H)$ (see, Eq. (13)) resulting in a relatively high standard deviation for the estimated parameter values. The ill-conditioning of the matrix $(H^T H)$ was verified by the high value of its condition number (about 10^5). It is worth observing that the high condition number of $(H^T H)$ derives from the DVL noise and not from the inadequate design of the input signal, since we designed the latter to be persistently exciting (Goodwin and Payne, 1977).

Since DVL uses the signal of a three-axis fluxgate magnetometer (among other sensors) to compute the instrument coordinate velocity, and since that this magnetometer is installed inside an ambience with high electromagnetic interferences, due to the vehicle eight thrusters, it is reasonable to expect that the velocity provided by DVL is not very precise.

The proposed identification procedure is based on the reduction of the parameter number of the model. Such reduction is carried out by determining the values of certain appropriate hydrodynamic parameters on the best of knowledge. This approach guarantees the parameter identifiability and has been used by Abkowitz (1980) and Hwang (1980) to determine the hydrodynamic parameters of a ship.

According to the proposed identification procedure, the estimated values of $k_{u|u|}$ and k_u listed in Table 3 (obtained from propelled tests) are introduced in Eq. (20) to identify parameters m_u , $X_{\dot{q}}$ and b . Then, Eq. (20) can be simplified to:

$$\ddot{u} = \alpha_1 \dot{q} + \alpha_2 R + \alpha_3, \quad (21)$$

where $\alpha_1 = -(X_{\dot{q}}/m_u)$, $\alpha_2 = (1/m_u)$, $\alpha_3 = -(b/m_u)$ and $R = -k_u u - k_{u|u|} |u| |u| + \tau_u$. R is a known force vector and the coefficients α_1 , α_2

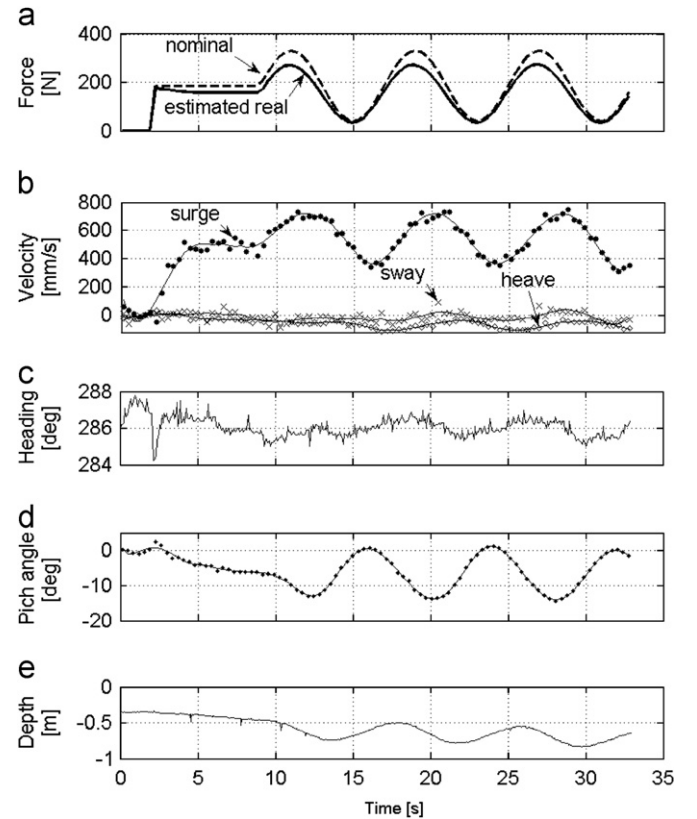


Fig. 10. Surge DOF: experimental data for a test with force input: $\tau_u^{nom} = 145.3 \sin(2 \times \pi \times 0.125t) + 185.7$ [N]. (a) Force input, (b) velocity, (c) heading, (d) pitch angle and (e) depth.

and α_3 are obtained by the parameter identification method outlined in Section 3.

A harmonic oscillatory motion was considered here to identify the hydrodynamic parameters of the vehicle. This motion was generated by applying a sinusoidal thrust through the thruster system. To investigate the effects of different thrust forces on the values of parameters m_u , $X_{\dot{q}}$ and b , four experiments with thrust profiles of the same amplitude and of different periods were conducted. The thrust profile is given by $\tau_u^{nom} = A \sin(2\pi t/P) + B$, where P is the period, A is the amplitude and B is the offset. The last two parameters were chosen so that no propeller inversion occurs. In all the experiments, the mean velocity of the vehicle (or offset of velocity) was 0.5 m/s. Fig. 10 shows the data collected during one of the variable velocity tests. In Fig. 10, the top plot shows a comparison of the nominal force (calculated using Eq. (1)) with the force actually applied upon the vehicle (calculated using Eq. (19)). The coupling between the surge and pitch motions is made clear in Fig. 10d and e.

Using all the experimental results, the estimated values for the coefficients of Eq. (21) are those listed in Table 4 and the corresponding hydrodynamic parameters are listed in Table 5. It is important to point out that the standard deviation of the estimated value of α_2 , in all the experiments, is smaller than 6.2%. Therefore, the estimated values for m_u are considered to be sufficiently accurate. The maximum standard deviation for the values estimated of α_1 and α_3 are of about 95% and 33%, respectively, except for the input of period 28 s, which was larger than 100%. The authors concluded that the relatively high level of uncertainty in the estimated values of α_1 and α_3 , or of $X_{\dot{q}}$ and b , are not important since the values of the hydrodynamic forces associated to these parameters are insignificant as compared with the thrust, inertia, and drag forces, as shown in Fig. 11.

Next, two non-dimensional parameters are introduced, the Reynolds number, Re , and the Keulegan–Carpenter number, KC , to characterize the viscous flow phenomena and, therefore, the drag force relevance on the vehicle dynamics. For the case of oscillating bodies completely submerged in water, these parameters are expressed according to Patel (1989) as

$$Re = \frac{U_m L_c}{\nu}, \quad KC = \frac{U_m T}{L_c} \quad (22)$$

where U_m is the maximum velocity of motion; T is the oscillation period; L_c is a characteristic length of the vehicle equal to $\forall^{1/3}$, where $\forall$ is the volume of the fluid displaced by the vehicle with value equal to 0.19 m³; and ν is the water kinematic viscosity. Besides, to verify whether the flow regime is dominated by drag or by inertia, the authors defined the ratio of the absolute values of the drag force and the inertia force as

$$\lambda = \frac{\sum_{k=1}^N |f_D(t_k)|}{\sum_{k=1}^N |f_I(t_k)|}, \quad (23)$$

where f_D and f_I are, respectively, the drag force $-k_u u - k_{u|u}|u|$ and inertia force $m_u \ddot{u}$, as given in Eq. (20).

Table 5 shows that m_u and the ratio of the drag force to the inertia force increase as the KC number increases. The fact is that by increasing the KC number, the flow regime is dominated by the viscous drag component. Furthermore, the sensitivity analysis of the hydrodynamic parameters showed that the added mass does not have much influence on the in-line force at high KC values, but a small change in the drag coefficient would significantly influence the in-line force (Avila, 2008).

The values of the drag and inertia parameters of LAURS obtained by testing the vehicle in a PMM are reported by Avila (2008). The increase of the surge virtual mass of LAURS with the KC number for $0.55 \leq KC \leq 5.46$ was also observed. At the top and bottom of Fig. 12, the values of the inertia coefficient C_m obtained from propelled tests (data denoted by solids circles), and using a PMM, are respectively shown. Here C_m is defined as being equal to $m_a/\rho \forall$, where m_a is the added mass induced by hydrodynamic pressure. Fig. 13 shows the same data of Fig. 12 but with coefficient C_m as a function of the KC number. According to Fig. 13, it is possible to conclude that the surge inertia coefficient of LAURS and therefore, the added mass, increases by increasing the KC number. For values of KC in the range of 0.55–39.82, the coefficient C_m increases in the range 0.4–3.5.

5.2. Performance assessment of the surge model under different experimental conditions

To assess the robustness of the surge model, the performance of one of the identified models under thrust profiles is evaluated in this section, which is different from that employed to obtain the parameters of the model. The model used in this assessment was the one obtained by applying the profile $\tau_u^{nom} = 145.3 \sin(2\pi t/28) + 185.7$ [N], as presented in Table 5, and is given by:

$$841.93\ddot{u} = 280.77\dot{q} - 82.3u - 309.07|u| + \tau_u + 0.05 \text{ [N]} \quad (24)$$

The simulation model was run for five cases of thrust force, see Fig. 14, and the nomenclature used to specify each case is A [N]/ B [N]/ P [s], where A , B and P are the parameters of the sinusoidal force $\tau_u^{nom} = A \sin(2\pi t/P) + B$. Case 1: 145.3/185.7/8, case 2: 145.3/185.7/12, case 3: 145.3/185.7/16, case 4: 145.3/185.7/28 and case 5: 63.7/104.0/8.

In Fig. 14, the simulated response using Eq. (24), dashed lines, is observed to closely agree with the experimentally measured

Table 5

Surge DOF: estimated values for virtual mass, coupling and bias terms of Eq. (20). Thrust Profile $\tau_u^{nom} = A \sin(2\pi t/P) + B$.

Force A (N)/ B (N)/ P (s)	Coefficients			Flow parameters		
	m_u (kg)	$X_{\dot{q}}$ (kg m)	b (N)	Re	KC	λ
145.3/185.7/8	694.1	192.3	10.8	4.13×10^5	10.02	2.3
145.3/185.7/12	750.4	122.3	9.05	4.22×10^5	15.37	3.1
145.3/185.7/16	816.8	161.3	9.96	4.48×10^5	21.75	3.3
145.3/185.7/28	841.9	−280.77	−0.05	4.68×10^5	39.82	6.1

Table 4

Surge DOF: estimated values for coefficients of Eq. (21). Thrust profile $\tau_u^{nom} = A \sin(2\pi t/P) + B$.

Force A (N)/ B (N)/ P (s)	α_1	α_2	α_3
145.3/185.7/8	-0.277 ± 0.045	$0.00144 \pm 3.46 \times 10^{-5}$	-0.01556 ± 0.0024
145.3/185.7/12	-0.163 ± 0.156	$0.00133 \pm 8.3 \times 10^{-5}$	-0.01206 ± 0.00407
145.3/185.7/16	-0.1975 ± 0.127	$0.00122 \pm 4.36 \times 10^{-5}$	-0.01219 ± 0.00205
145.3/185.7/28	0.3335 ± 0.223	$0.00119 \pm 6.03 \times 10^{-5}$	0.000059 ± 0.00168

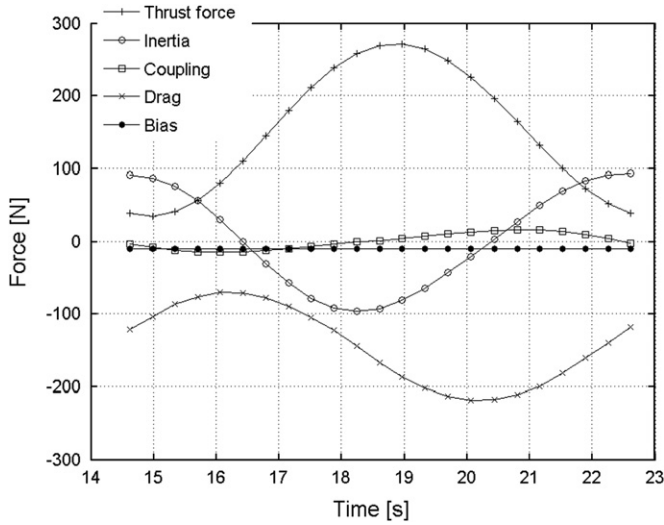


Fig. 11. Surge DOF: force components acting upon the vehicle in test with force input: $\tau_u^{nom} = 145.3 \sin(2 \times \pi \times 0.125t) + 185.7$ [N].

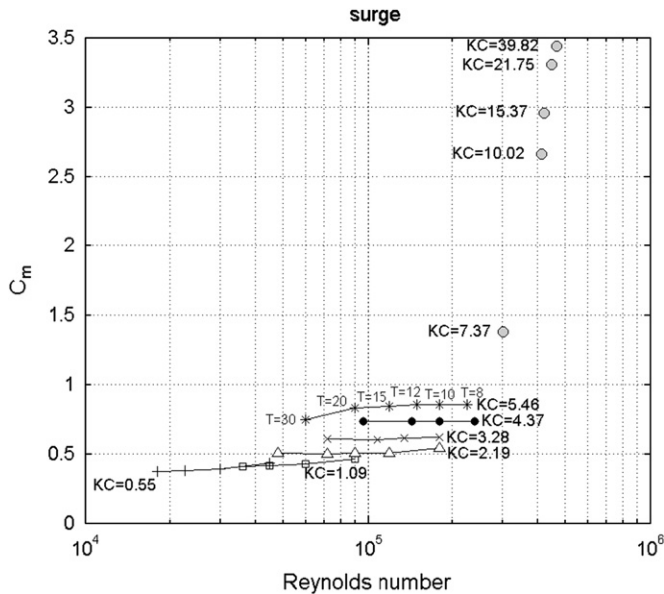


Fig. 12. Surge DOF: values of inertia coefficient of the vehicle LAURS, in propelled tests (top) and using a PMM (bottom).

response under different thrust profiles. In case 5, simulated and measured responses matched very well, even when the amplitude of the input force applied to the vehicle is smaller than the amplitude of the input force used to identify the parameter of the model. In each case, the simulated model response, u_{simul} , is compared to the experimentally measured response, $u_{measured}$, and the simulation error, E , between the simulated and measured responses is calculated as follows:

$$E = \text{mean}(|u_{measured} - u_{simul}|),$$

$$\sigma = \text{std}(|u_{measured} - u_{simul}|) \quad (25)$$

Table 6 shows the statistical errors. The mean value of E is observed to increase by decreasing KC. Based on the results in Table 6 and by visual inspection, the authors concluded that the model given by Eq. (24) has an acceptable performance for variations of virtual mass below 20% of the original value, or, for KC numbers larger than 10.02.

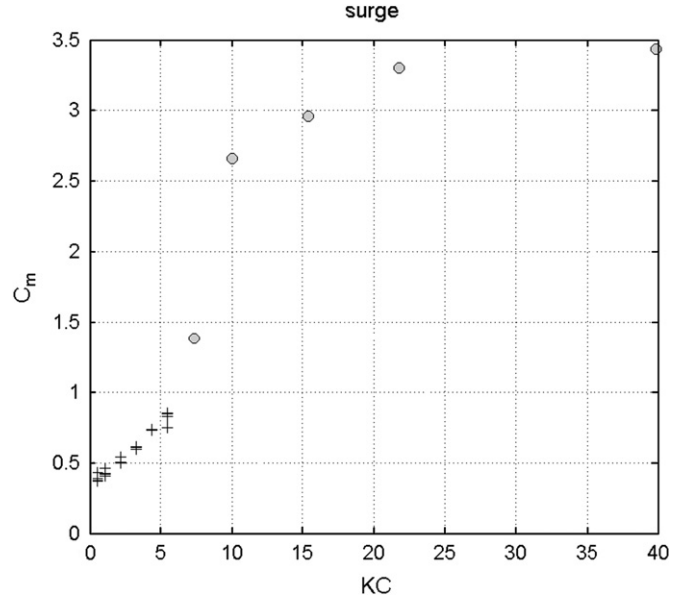


Fig. 13. Surge DOF: values of inertia coefficient of the vehicle LAURS, in propelled tests (denoted by solid circles) and using a PMM (denoted by plus symbol).

5.3. Yaw model identification

Multiple tests in the yaw DOF were conducted with constant and sinusoidal torque profiles to determine the yaw hydrodynamic parameters of the LAURS. During the experiments, thrusters T5 and T8 were used to apply positive torques and thrusters T2 and T6 were used to apply negative torques. The yaw dynamic model is given by

$$I_r \dot{r} = -k_{r|r} r |r| - k_r r + \tau_r - b, \quad (26)$$

where r and $\dot{r}$ are the yaw angular velocity and acceleration, respectively; τ_r is the yaw torque applied; $k_{r|r}$ and k_r are the quadratic and linear drag parameters, I_r is the virtual inertia and b is the bias of the model. After some manipulation, Eq. (26) can be written as

$$\dot{r} = \alpha_1 R + \alpha_2, \quad (27)$$

where $\alpha_1 = (1/I_r)$, $\alpha_2 = (-b/I_r)$ and $R = -k_{r|r} r |r| - k_r r + \tau_r$.

Fig. 15 shows the experimental data obtained from constant velocity tests and Table 7 lists the values of the drag parameters obtained by fitting the experimental data with the steady-state yaw model. This model was obtained by making $\dot{r} = 0$ in Eq. (26).

Fig. 16 shows the data collected in one of the experiments with a sinusoidal torque profile. Table 8 lists the estimated values of the coefficients of Eq. (27) for four different torque profiles, which were obtained by using the identification method outlined in Section 3. Table 9 presents the estimated values of the virtual inertia. The values of I_r are observed not to change significantly with the torque profiles considered, in contrast with the value of the virtual mass in surge. According to the values of λ , it can be deduced that the relevance of the torque due to the drag on the inertia torque increases by increasing the amplitude of the yaw oscillation. This result is similar to that obtained for the tests in surge, as Table 5 shows.

5.4. Zig-zag maneuvers

This section presents the dynamic model for the motion of the LAURS in the horizontal plane, composed by three coupled differential equations for the surge, sway and yaw DOFs. Next, the values of hydrodynamic parameters that correspond to the

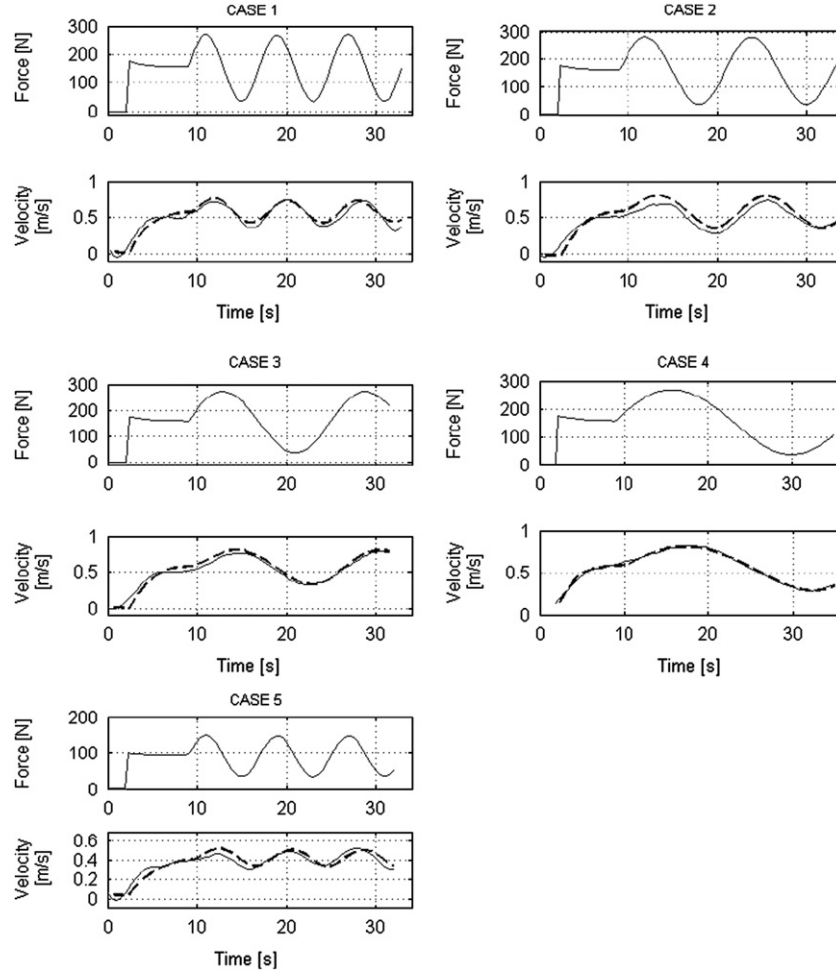


Fig. 14. Surge DOF: comparison of performance of Eq. (24) under different sinusoidal thrust profiles.

Table 6

Surge DOF: Error parameter for the simulations shown in Fig. 13. $\tau_u^{nom} = A \sin(2\pi t/P) + B$.

Force $A(N)/B(N)/P(s)$	KC	E
145.3/185.7/28	39.82	0.01355 ± 0.00854
145.3/185.7/16	21.75	0.03714 ± 0.02530
145.3/185.7/12	15.37	0.06317 ± 0.03585
145.3/185.7/8	10.02	0.04436 ± 0.03352
63.7/104.0/8	7.37	0.03595 ± 0.02621

yaw model equation are reported. The yaw model was chosen because of the high dynamics in the yaw axis of the vehicle during zig-zag maneuvers. A comparison between the values of the parameters obtained from zig-zag maneuvers and 1-DOF experimental trials is made. Finally, the values of the parameters of the yaw model obtained by using the LS method directly (without integration of the differential equation) are compared with the values obtained using the proposed identification procedure.

The equations for the zig-zag steering motion of the LAURS are obtained from the general motion equations of a ROV, which were simplified by considering an intrinsic stable motion in roll and pitch for the vehicle, given by Fossen (1994) and Caccia and Veruggio (2000), as follows:

$$m_u \dot{u} = m_v vr - k_u u - k_{|u|} |u| + \tau_u, \quad (28a)$$

$$m_v \dot{v} = -m_u ur - k_v v - k_{|v|} |v| + \tau_v, \quad (28b)$$

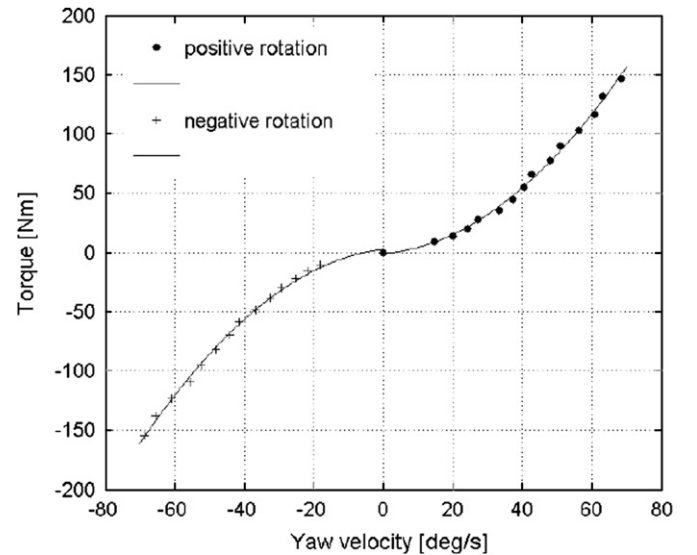


Fig. 15. Yaw DOF: experimental data for tests of constant angular velocity.

$$I_r \dot{r} = -(m_v - m_u)uv - k_r r - k_{|r|} r |r| + \tau_r, \quad (28c)$$

where variables are defined again for the convenience of the reader: u , v and r are the surge, sway and yaw velocities, respectively; m_u and m_v are the surge and sway virtual mass,

Table 7
Yaw DOF: drag parameters of the LAURS.

Sense of rotation	$k_{r r }$ (kg m ²)	k_r (kg m ² /(rad s))	b (N m)
Clockwise	95.32 ± 5.93	12.78 ± 7.79	−1.10 ± 2.35
Anticlockwise	94.72 ± 5.31	18.21 ± 7.15	−2.53 ± 2.24

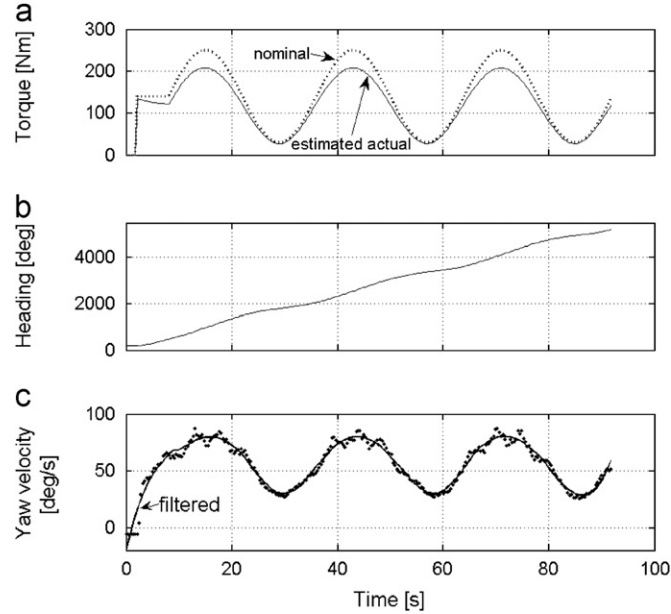


Fig. 16. Yaw DOF, experimental data for variable velocity test with torque profile $\tau_r^{nom} = 110.1 \sin(2\pi t/28) + 139.2$ [Nm]

respectively, and I_r is the yaw virtual inertia; τ_u and τ_v are the forces applied by the thrusters in the surge and sway directions and τ_r is the applied yaw torque. The term $(m_v - m_u)uv$ of Eq. (28c) is the Munk moment, which is always destabilizing, since it acts to turn the vehicle perpendicularly to the flow (Triantafyllou and Hover, 2003). The parameter $(m_v - m_u)$ is denoted by k_{uv} here.

The zig-zag maneuvers were carried out by simultaneously applying a constant force in surge and a constant torque in yaw, so that the sense of application of the torque was changed cyclically from positive to negative (or vice versa). The yaw angular displacement was controlled to avoid saturation, defined as $\Delta\psi_{max}$. During the experiments, the surge force was applied by thrusters T5 and T6, while the yaw torque was applied by thrusters T2 and T8; see Fig. 5.

Following the proposed identification procedure, Eq. (28c) is written as:

$$\dot{r} = \alpha_1 uv + \alpha_2 R + \alpha_3, \quad (29)$$

where $\alpha_1 = -(k_{uv}/I_r)$, $\alpha_2 = (1/I_r)$, $\alpha_3 = -(b/I_r)$ and $R = -k_r r - k_{r|r|}r|r| + \tau_r$. Vector R was calculated using the values of the coefficients k_r and $k_{r|r|}$ listed in Table 7. The following expression was used to calculate the input torque:

$$\tau_r = \eta_{adv} \tau_r^{nom},$$

where η_{adv} is given by Eq. (17).

In this case, two different experiments were conducted. Fig. 17 shows the data acquired during the experiment with surge velocity equal to 0.457 m/s. The parameter identification method explained in Section 3 was used to obtain the coefficients of Eq. (29). Subsequently, using the values of these coefficients, numerical simulations using the fourth order Runge–Kutta ODE integration algorithm were carried out. Fig. 18b compares

Table 8

Yaw DOF: estimated values of the coefficients of Eq. (27) with torque profile $\tau_r^{nom} = A \sin(2\pi t/P) + B$.

Torque A(N m)/B(N m)/P(s)	α_1	α_2
43.86/72.95/8	0.004655 ± 0.0003751	−0.013321 ± 0.009397
43.86/72.95/20	0.004176 ± 0.0001018	−0.018437 ± 0.001246
43.86/72.95/28	0.004837 ± 0.0001488	−0.023214 ± 0.001346
110.09/139.18/28	0.004909 ± 0.000137	−0.027480 ± 0.002101

Table 9

Yaw DOF: estimated values of the virtual inertia with torque profile $\tau_r^{nom} = A \sin(2\pi t/P) + B$.

Torque A(N m)/B(N m)/P(s)	I_r (kg m ² /rad)	b (N m)	λ
43.86/72.95/8	214.82	2.86	2.74
43.86/72.95/20	239.44	4.41	5.85
43.86/72.95/28	206.72	4.80	10.54
110.09/139.18/28	203.69	5.60	8.76

numerically simulated and experimentally measured responses, in which the gyroscope was used. The simulated response is observed to match the measured response very well. Table 10 shows the estimated values of the coefficients of Eq. (29) and the values of the virtual inertia, coupling coefficient and bias. The virtual inertia I_r obtained from zig-zag maneuvers is observed to be on average 20% smaller than that obtained in the uncoupled yaw tests; see Table 9. The difference in I_r obtained from pure yaw (1-DOF) and zig-zag (3-DOFs) tests was expected, since the water flow produced by the pure yaw is totally developed, whereas in zig-zag motion it is partially developed, due to the constrained yaw angular displacement.

It is also observed that k_{uv} is a function of the vehicle surge velocity and, in contrast to parameter X_q of Eq. (20), k_{uv} has considerable influence on the vehicle dynamic behavior, as shows the coupling torque associated to k_{uv} in Fig. 18c.

At the beginning of this section, the values of the parameters obtained using the identification procedure proposed were presented. For comparison of results, values of the parameters obtained by identifying all the parameters of the model at once using a single test are presented. It is important to point out that the same experimental data was used in both procedures. The model given by Eq. (28c) can be written as:

$$\dot{r} = \alpha_1 uv + \alpha_2 r + \alpha_3 r|r| + \alpha_4 \tau_r + \alpha_5, \quad (30)$$

where $\alpha_1 = -(k_{uv}/I_r)$, $\alpha_2 = -(k_r/I_r)$, $\alpha_3 = -(k_{r|r|}/I_r)$, $\alpha_4 = (1/I_r)$ and $\alpha_5 = -(b/I_r)$. Considering $\dot{r}$ to be known and obtained by numerical differentiation of velocity r , the LS method was directly applied to obtain the coefficients of Eq. (30). Table 11 presents the estimated values of these coefficients, in which the corresponding hydrodynamic parameters are also listed. It is observed that the values of k_r and $k_{r|r|}$ are absurd as compared with the values obtained from the pure yaw tests of constant velocity (data of reference); see Table 7. $k_{r|r|}$ Take negative values, which it not possible because of the dissipative nature of the hydrodynamic damping. Also, the values of I_r are observed to be about 50% of the values obtained from pure yaw oscillatory tests; see Table 9.

Moreover, considering the absolute values, results for k_{uv} are small as compared to the values obtained using the identification procedure proposed; see Table 10. Although the values of the hydrodynamic parameters of Eq. (28c), obtained by applying the LS method directly, are very different from the values obtained using the identification procedure proposed (based on the differential equation integration) the measured and simulated responses are quite similar. The authors conclude that there is

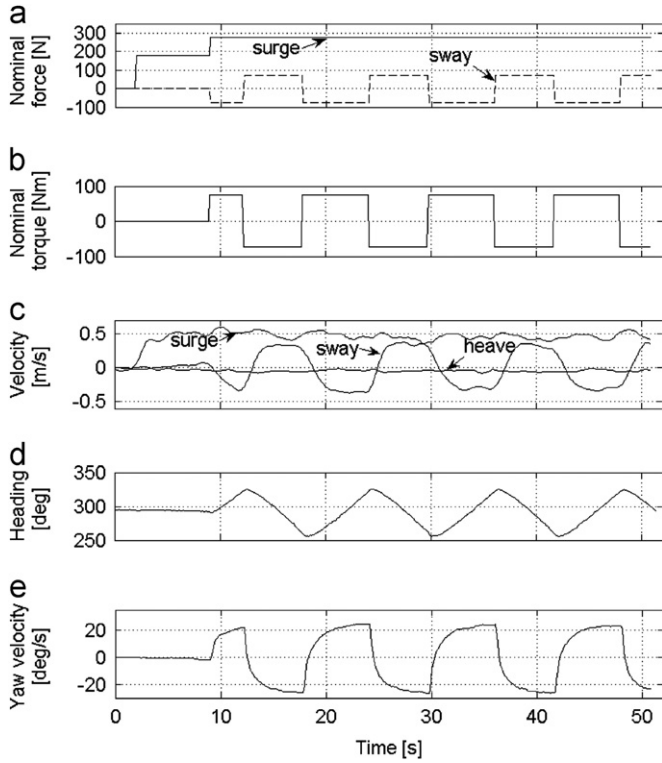


Fig. 17. Zig-zag maneuver: experimental data.

qa problem of parameter identifiability, since there is not one single solution to the parameters that describes the input–output relationship. Techniques of parallel processing (Hwang, 1980) as well as artificial intelligence methods have been used to solve the problem of parameters identifiability in case of ships and underwater vehicles, respectively, but they are not addressed here, because this was considered out of the scope of the present work.

6. Conclusions

An identification procedure for hydrodynamic parameters for coupled and uncoupled mathematical models for open-frame underwater vehicles is presented. The focus is to determine “actual values” for the hydrodynamic parameters and also to guarantee that simulated and measured responses match, which is different from most of the related publications that only focus on the latter case. The identification procedure may be summarized as follows. First, the efficiency coefficients of the system of thrusters operating close to the vehicle structure are identified through experiments. The efficiency coefficients, η_{inter} and η_{adv} defined in Eq. (16), are used to compute the torque/force actually applied upon the vehicle, taking into account the loss of trust due to the thruster–thruster and thruster–hull interactions and to the velocity of advance of the vehicle. Subsequently, drag parameters are identified from constant velocity tests and, finally, using the estimated values of the drag parameters, the inertia and coupling parameters from tests of variable velocity are identified experimentally. An advantage of this identification procedure is that instruments for acceleration measurement are not required. Since the parameter estimation method considered in this work is the least-squares algorithm applied to the integral form of the system dynamic equations, the need for acceleration measurement is eliminated. The proposed procedure was used to identify the hydrodynamic parameters of a 1-DOF model for surge and yaw

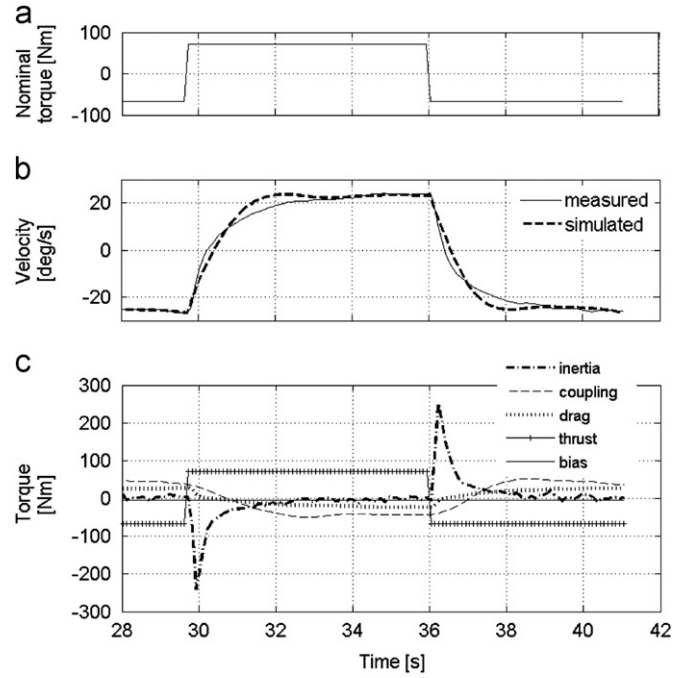


Fig. 18. Zig-zag maneuver. (a) Nominal torque, (b) simulated and measured yaw velocities and (c) total torque components.

Table 10

Zig-zag Maneuvers: hydrodynamic parameters estimated values for yaw model.

τ_r (Nm)	u_0 (m/s)	$\Delta\psi_{max}$ (deg.)	α_1	α_2	α_3
70.76	0.457	60	2.119 ± 0.147	0.0064 ± 0.0003	-0.0324 ± 0.014
			k_{uv} (kg)	I_r (kg m ² /rad)	b (Nm)
			–329.22	155.40	5.03
67.80	0.714	50	0.9307 ± 0.0689	0.0054 ± 0.0003	-0.0138 ± 0.0137
			k_{uv}	I_r	b
			–172.76	185.63	2.56

motion. Moreover, the procedure was also used to identify the 3-DOF model for the yaw dynamic in a horizontal zig-zag motion.

It was shown that the thruster–thruster and thruster–hull interactions and the effect of the advance velocity on the efficiency of the thrusters may be extremely important in the dynamics of open-frame ROVs and must thus be taken explicitly into account. Experimental data acquired with the LAURS vehicle in bollard-pull conditions show that the thruster efficiency loss due to thruster–thruster and thruster–hull interactions is about 10% for surge, sway and heave directions, compared to the open-water thrust thruster model, see Table 2. As information, Caccia et al. (2000) report a thruster efficiency loss of 44% during the heave motion of an open-frame ROV. They determined the overall efficiency coefficient (η_{inter}) and the hydrodynamic parameters of the model using the data of one single test. In contrast with the work reported by Caccia et al. (2000), in the present work, η_{inter} was obtained exclusively from bollard-pull tests, due to precision considerations. In addition, the number of parameters to be determined was reduced, because η_{inter} is known and the uncertainty in the estimated values of the hydrodynamic parameters also decreases (Hwang, 1980).

The identification procedure proposed here has a good performance, even when sensors with low sampling rate and acquired signals with relatively high noise level are used. The values of the parameters obtained from propelled tests were shown to be consistent with the values obtained by towing the vehicle in

Table 11

Zig-Zag Maneuvers: hydrodynamic parameters—estimated values for yaw model—Eq. (30).

τ_r (Nm)	u_0 (m/s)	$\Delta\psi_{max}$ (deg)	α_1	α_2	α_3	α_4	α_5
70.76	0.457	60	0.4161 ± 0.2735 k_{uv} (kg) –49.36	-2.4815 ± 0.2157 k_r (kg m ² /(rad s)) 294.33	3.1052 ± 0.4766 $k_{r r }$ (kg m ²) –368.31	0.0084 ± 0.0003 I_r (kg m ² /rad) 118.61	-0.038 ± 0.0098 b (Nm) 4.51
67.80	0.714	50	0.1427 ± 0.0943 k_{uv} –19.30	-2.2849 ± 0.1708 k_r 308.90	3.0504 ± 0.4499 $k_{r r }$ –412.39	0.0074 ± 0.0002 I_r 135.19	-0.0374 ± 0.0078 b 5.06

a water tank, in spite of the fact that, the velocity sensor used in the identification, the DVL, has a sampling rate of 3 Hz and the acquired velocity signal did not present good behavior. Fig. 9 shows that the surge drag curves obtained from tests with propulsion and by towing the vehicle match very well, thus showing the potential of the procedure proposed.

The analysis of results allows concluding that the surge virtual mass, m_{uv} , and therefore, the inertia coefficient, C_m , increases by increasing the KC number (or the amplitude of oscillation), when C_m takes values between 0.4 and 3.5 and KC between 0.55 and 39.82, as shown in Fig. 13. Moreover, in the same range as KC, the total force acting upon the vehicle is dominated by the viscous drag component, as show the values of λ in Table 5. This conclusion contrasts with most of the related publications which do not present results of variation of the hydrodynamic coefficients with the KC number.

It is worth pointing out that in all experiments the identified parameter model response closely agrees with the experimentally measured response, which shows a very robust identification procedure. The effect of different thrust profiles on the dynamic performance of the identified parameter models was also analyzed, and the conclusion is that the models obtained with higher KC numbers work well for the self high values of KC and for values of KC around the ones. In fact, it was verified that the surge model obtained with $KC=39.82$ has a good performance until the experimental condition given by $KC=10.02$. In other words, a reduction in value of the virtual mass equivalent to the 20% of the reference value is permitted.

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